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**FACULTY OF SCIENCE & TECHNOLOGY  
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# 1. Project Overview

This project presents a comprehensive exploratory data analysis (EDA) of the PlantVillage Dataset — a widely-used agricultural image dataset for plant disease detection. The analysis was conducted using Python with Pandas, NumPy, and Matplotlib to uncover insights about crop health distribution, disease variety, and dataset structure.

The PlantVillage Dataset contains over 54,000 labeled images of plant leaves organized into 38 categories across 14 crop species. Each category represents either a specific disease or a healthy state for a given crop.

## 1.1 Analytical Questions

Three core questions guided this investigation:

- Crop Vulnerability: Which crop species exhibit the widest variety of diseases, indicating higher susceptibility?
- Health Distribution: What is the overall ratio of healthy vs. diseased plants, and how does this vary across crops?
- Data Quality & Consistency: Are there statistical anomalies in the dataset structure (e.g., filename patterns) that could indicate collection inconsistencies?

## 1.2 Why These Questions Matter

Understanding which crops are most disease-prone helps agricultural researchers prioritize resources. Analyzing health distribution ensures that any machine learning model trained on this data reflects real-world agricultural conditions. Data quality checks ensure the reliability of the dataset for downstream use.

# 2. Dataset Description

The PlantVillage Dataset is a publicly available, open-source collection of plant leaf images originally compiled for plant disease classification research. It is widely used as a benchmark dataset in computer vision and precision agriculture.

## 2.1 Dataset Summary

| Property                      | Value    |
|-------------------------------|----------|
| Total Images                  | 54,305   |
| Unique Disease/Health Classes | 38       |
| Crop Species                  | 14       |
| Healthy Image Count           | 15,084   |
| Diseased Image Count          | 39,221   |
| Missing Values                | None (0) |

|                   |                               |
|-------------------|-------------------------------|
| Duplicate Entries | None (0)                      |
| Data Source       | Kaggle / PlantVillage Project |
| Image Format      | .jpg / .png                   |
| Structure         | Folder-based labeling         |

The dataset was loaded by traversing the directory structure where each sub-folder name encodes both the crop species and its health condition (e.g., Apple\_\_\_Apple\_scab or Tomato\_\_\_healthy). The two original columns in the constructed DataFrame were Image\_Name and Disease\_Type.

### 3. Data Cleaning and Preparation

Three systematic cleaning steps were applied to ensure data integrity before analysis:

#### Step 1 — Handling Duplicates

Problem: The dataset may contain redundant entries where the same image is listed multiple times.

Action: Applied `drop_duplicates()` to remove all duplicate rows.

Result: Zero duplicates found — 54,305 rows retained.

Justification: Duplicate removal ensures that statistical distributions are not biased by repeated data points, which would artificially inflate counts for certain categories.

#### Step 2 — Column Name Standardization

Problem: Original column names used mixed-case with spaces, which can cause attribute access errors.

Action: Standardized all column names to lowercase snake\_case (image\_name, disease\_type).

Justification: Consistent naming follows Python PEP 8 conventions, making code more readable and preventing syntax errors during filtering and grouping operations.

#### Step 3 — Text Data Validation and Stripping

Problem: Categorical strings may contain leading/trailing whitespace due to folder naming inconsistencies.

Action: Applied `str.strip()` to all disease\_type entries.

Justification: Without stripping, categories like 'Apple' and 'Apple ' would be treated as separate groups, causing incorrect aggregations.

### 4. Feature Engineering

Three new derived columns were created to enable deeper analysis:

| New Column    | Derivation Method                                   | Purpose                                    |
|---------------|-----------------------------------------------------|--------------------------------------------|
| crop          | Split disease_type on '___' or ' ', take first part | Enables crop-level subgroup comparisons    |
| health_status | Checks if 'healthy' appears in disease_type string  | Enables diseased vs. healthy comparison    |
| name_length   | Measures character length of image_name string      | Enables NumPy statistical outlier analysis |

The crop and health\_status columns were critical for answering the project's analytical questions. The name\_length column provided a numerical proxy for data consistency checks using NumPy.

## 5. Data Analysis

### 5.1 Subgroup Comparison: Crop Health Status

A groupby operation on crop and health\_status revealed major imbalances across species. Key observations:

- Tomato has the largest total image count (18,160) with only 1,591 healthy images — a highly imbalanced class.
- Blueberry, Raspberry, and Soybean contain only healthy samples with no disease categories.
- Orange and Squash contain only diseased samples with no healthy baseline.
- Apple has a relatively balanced split: 1,526 diseased vs. 1,645 healthy.

| Crop         | Diseased | Healthy | Total  |
|--------------|----------|---------|--------|
| Tomato       | 16,569   | 1,591   | 18,160 |
| Orange       | 5,507    | 0       | 5,507  |
| Soybean      | 0        | 5,090   | 5,090  |
| Grape        | 3,639    | 423     | 4,062  |
| Corn (Maize) | 2,690    | 1,162   | 3,852  |
| Apple        | 1,526    | 1,645   | 3,171  |
| Peach        | 2,297    | 360     | 2,657  |
| Pepper, Bell | 997      | 1,478   | 2,475  |
| Potato       | 2,000    | 152     | 2,152  |
| Cherry       | 1,052    | 854     | 1,906  |

### 5.2 Ranking: Disease Variety by Crop

Tomato leads with 10 unique disease categories, followed by Apple and Grape (4 each), and Corn (4). This ranking directly addresses the Crop Vulnerability question.

| Crop                                              | Unique Disease Types |
|---------------------------------------------------|----------------------|
| Tomato                                            | 10                   |
| Apple                                             | 4                    |
| Grape                                             | 4                    |
| Corn (Maize)                                      | 4                    |
| Potato                                            | 3                    |
| Cherry                                            | 2                    |
| Strawberry                                        | 2                    |
| Peach                                             | 2                    |
| Pepper, Bell                                      | 2                    |
| Blueberry / Raspberry / Squash / Soybean / Orange | 1 each               |

### 5.3 Overall Health Distribution

The frequency analysis of health\_status showed:

- Diseased: 39,221 images — 72.22% of total dataset
- Healthy: 15,084 images — 27.78% of total dataset

This significant imbalance is critical for any future machine learning implementation. Models trained on this dataset will have a natural bias toward disease detection unless class balancing techniques are applied.

### 5.4 NumPy Statistical Computation

Custom NumPy computations were applied to the name\_length feature:

- Mean Filename Length: 56.53 characters
- Standard Deviation: 6.98 characters

The low standard deviation relative to the mean indicates a highly standardized naming convention across all 54,305 images, suggesting consistent data collection practices.

### 5.5 Outlier / Anomaly Analysis (Z-Score)

Z-scores were calculated for all filename lengths. Entries with  $|Z| > 3$  were flagged as anomalies:

- Anomalous entries detected: 1,193 (approximately 2.2% of the dataset)

These entries have significantly longer or shorter filenames than the mean. While this doesn't necessarily indicate corrupted data, it warrants further inspection during preprocessing for machine learning workflows. The anomalies likely correspond to images with non-standard source codes embedded in their filenames.

### 5.6 Relationship Analysis: Image Volume vs. Disease Variety

A correlation-style comparison was performed between total images per crop and unique disease types per crop:

- Tomato: 18,160 images, 10 unique diseases — strong volume-variety relationship
- Soybean: 5,090 images, 1 unique class (healthy) — high volume but low variety
- Orange: 5,507 images, 1 unique class — similar to Soybean

The scatter plot analysis revealed that higher image volume does NOT always correspond to higher disease variety. Dataset collection choices, not just image count, drive categorical diversity.

## 6. Visualizations

Four key charts were produced using Matplotlib. Screenshots of these charts would be embedded here in the final submitted report.

| Chart | Type         | Title                                      | Key Insight                                                                                         |
|-------|--------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 1     | Bar Chart    | Distribution of Images Across Crops        | Tomato dominates (18K+). Orange, Soybean also large. Raspberry has fewest images.                   |
| 2     | Pie Chart    | Healthy vs. Diseased Distribution          | 72.2% diseased, 27.8% healthy — severe class imbalance.                                             |
| 3     | Histogram    | Frequency Distribution of Filename Lengths | Tight distribution around mean (56.53). Red dashed line marks mean. Low spread = consistent naming. |
| 4     | Scatter Plot | Total Images vs. Unique Disease Varieties  | No strong linear relationship. Tomato is an outlier. Volume doesn't guarantee diversity.            |

## 7. Key Findings

Five evidence-based insights emerged from the analysis:

### Finding 1: Tomato is the Most Vulnerable Crop

Tomato exhibits 10 unique disease categories — more than twice any other crop in the dataset. This indicates either a higher real-world susceptibility, more targeted research attention, or both. Agricultural stakeholders should prioritize Tomato disease monitoring in precision farming systems.

### Finding 2: Dataset is Heavily Biased Toward Diseased Samples

Over 72% of all images are classified as diseased. This skew means any classifier trained on this data will have significantly more exposure to diseased patterns than healthy ones. Class-balancing (oversampling, undersampling, or weighted loss functions) is essential before model training.

### Finding 3: Several Crops Lack a Healthy Baseline

Orange and Squash have zero healthy images in the dataset. This is a critical limitation — without a healthy reference class, these crops cannot be used in binary healthy/diseased classification tasks without additional data collection.

#### **Finding 4: Data Collection was Highly Standardized**

The NumPy statistical analysis of filename lengths revealed a mean of 56.53 with a standard deviation of just 6.98. This low variance confirms consistent, automated data collection with standardized naming conventions — a positive indicator for dataset quality.

#### **Finding 5: Image Volume Does Not Predict Disease Diversity**

The scatter plot showed no strong positive correlation between the number of images collected for a crop and the number of unique disease types represented. This finding is important for dataset curation: simply collecting more images of one crop does not automatically improve class coverage.

## **8. Limitations**

The following limitations were identified during the analysis:

- **Categorical Depth Only:** The dataset is folder-name based. Any human labeling errors during initial collection would propagate directly into analysis without detection.
- **No Environmental Metadata:** Geographic location, climate, season, or weather data are absent. This prevents analysis of environmental correlations with disease outbreaks.
- **Limited Numerical Features:** As an image dataset, numerical analysis was constrained to proxy measures (filename length). Deeper numerical insights require pixel-level or EXIF metadata extraction.
- **Class Imbalance:** The heavy skew toward Tomato and toward diseased samples means findings may not be equally generalizable to all species or health conditions.
- **Correlation vs. Causation:** Statistical relationships found (e.g., disease variety per crop) reflect dataset composition choices, not necessarily real-world agricultural patterns.

## **9. Conclusion**

This exploratory data analysis of the PlantVillage Dataset provided valuable insights into its structure, balance, and quality. Using Pandas for data manipulation, NumPy for statistical computation, and Matplotlib for visualization, we answered all three analytical questions:

- Tomato is the most disease-diverse crop (10 unique disease types), followed by Apple, Grape, and Corn.
- The dataset is dominated by diseased samples (72.2%), with a significant class imbalance that must be addressed in any ML application.
- Filename length analysis confirmed highly consistent data collection, though 2.2% of entries showed statistical anomalies worth investigating.

The dataset is well-suited for plant disease classification tasks but requires preprocessing — particularly class balancing and verification of crops lacking healthy baselines — before deployment in production machine learning pipelines.

## 10. References

- PlantVillage Dataset: Available on Kaggle — <https://www.kaggle.com/datasets/abdallahalidev/plantvillage-dataset>
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